Topological Structures and Signals

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Intelligent Systems Department

May 15, 2026

Beyond *Pairwise* Data Relationships

- Graphs are a limited example of topological space

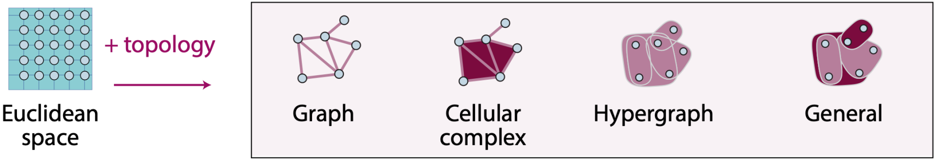

fig.credit: Sanborn. al

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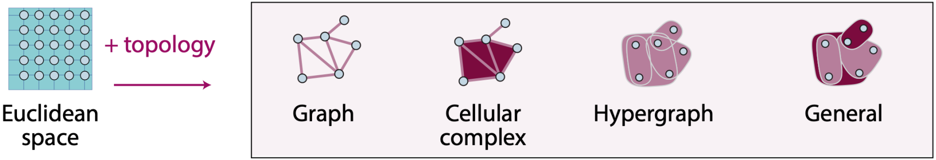

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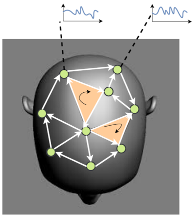
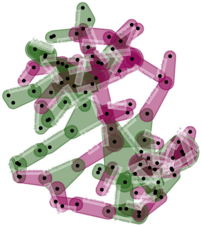
- Graphs encode only pairwise relations
  - ⇒ Several brain regions activate together
  - ⇒ Multiple co-collaborators in a single project



Complex systems



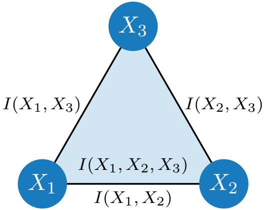
Coauthorship network

Beyond *Pairwise* Data Relationships

- Motivation for encoding multiway relations

- Information-theoretic perspective
  - $\Rightarrow$ Pairwise mutual information:

$$I(X_1, X_2) = H(X_1) + H(X_2) - H(X_1, X_2)$$



Beyond *Pairwise* Data Relationships

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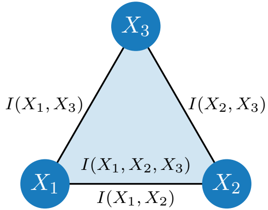

- Information-theoretic perspective

  ⇒ Pairwise mutual information:

$$I(X_1, X_2) = H(X_1) + H(X_2) - H(X_1, X_2)$$

- Triple-wise mutual information:

$$I(X_1, X_2, X_3) = H(X_1) + H(X_2) + H(X_3) - H(X_1, X_2) - H(X_2, X_3) - H(X_1, X_3) + \underbrace{H(X_1, X_2, X_3)}_{\text{tot. entropy of 3 var.}}$$

Mutual information between 3 random variables cannot be reduced to pairwise mutual information

- We need higher-order or multiway structures

Examples of Multiway relationships

- Brain Networks
  - ⇒ Interactions are beyond pairwise
  - ⇒ Complex tasks require higher levels of coordination involving more brain regions.

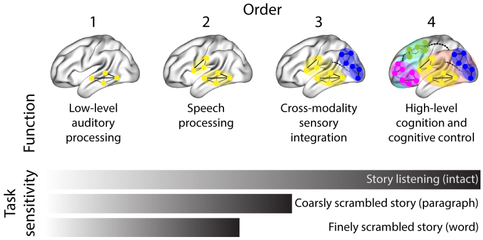

fig. credit Owen et al.,

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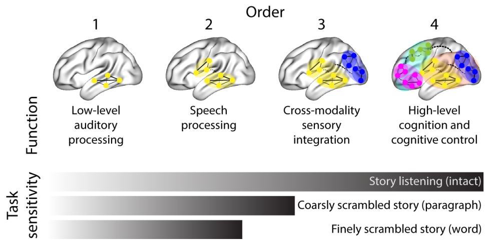

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- Contagion Dynamics
  - ⇒ Diseases do not only spread when pairs of people interact
  - ⇒ Contagion depends not only on how many neighbors are infected, but on how they are grouped into interacting clusters.

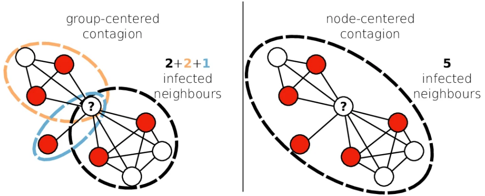


fig. credit Burgio et al.,

Examples of Multiway Relationships

- Protein Interaction Networks

  ⇒ Groups of proteins interact together rather than through separate pairwise links to determine functional outcomes.

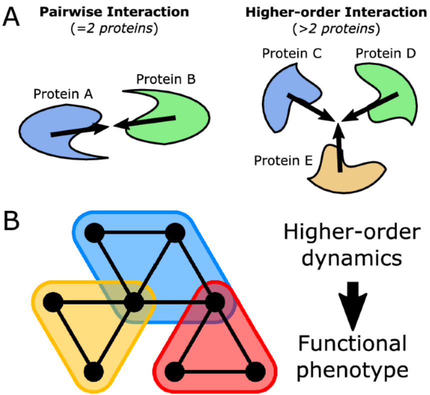


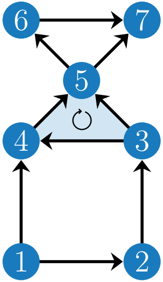
fig. credit Murgas et al.,

Topological Data Relations

Definition. A topological space is a set of elements along with a set of neighboring properties expressing closeness

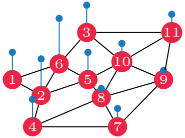
- Example: $\tau = \{1, \ldots, 7, \{1,2\}, \ldots, \{6,7\}, \{3,4,5\}\}$

  $\Rightarrow$ Triplet $\{5,6,7\}$ and quadruplet $\{1,2,3,4\}$ are not part of the topological object



Beyond *Node-centric* Signals

- Graph signal processing, graph neural networks and alternatives are principled approaches to process signals defined on the nodes



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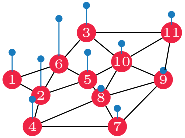
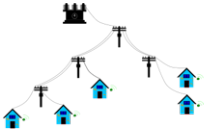
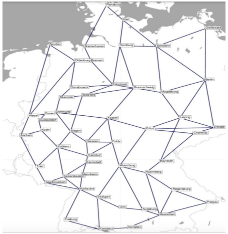


- Graph signal processing, graph neural networks and alternatives are principled approaches to process signals defined on the nodes
  - Increasing demand to process signals on tuples such as flows
  - Joint processing of node + edge signals in critical infrastructure networks

Water networks



Power networks



Communication networks

Signal over Topological Spaces

Definition. A signal defined over a topological space $\mathcal{S}$ is a mapping from the elements of the topological space to the real domain

$$f : \mathcal{S} \to \mathbb{R}$$

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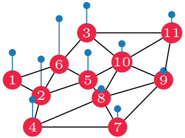
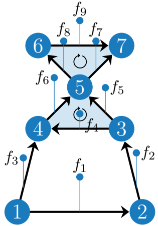


- Ex. graph signal defined over the nodes
- Ex. edge signals defined over the edges (pairs of nodes)



Outline: What is This Lecture About?

- How to represent topological objects and signals?

$\Rightarrow$ Machine learning tasks with topologies

$\Rightarrow$ Simplicial and cell complexes

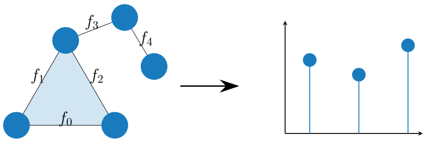
$\Rightarrow$ Hodge Laplacians

$\Rightarrow$ Signal mappings

Machine Learning with Topologies

Core objective: learn representations from topological structures and signals

- Sparse representations
  - ⇒ Find a sparse representation of a topological signal
  - ⇒ Ex. identify dictionary atoms to promote sparsity
- Filtering
  - ⇒ Discriminate spectral signal components



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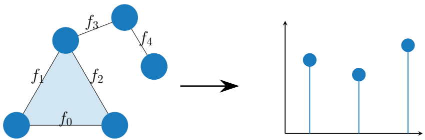
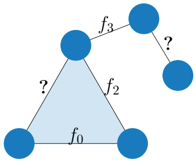


- Filtering
  - ⇒ Discriminate spectral signal components

- Active learning and regression
  - ⇒ Sample a signal on a few simplexes and reconstruct missing values
  - ⇒ Ex. recover observations of faulty sensors



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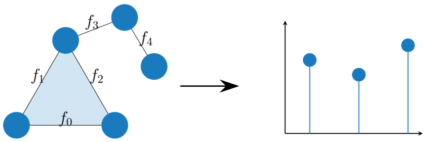

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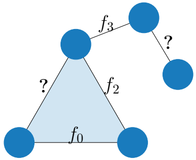

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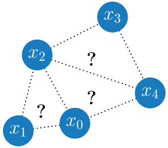

- Topology inference
  - ⇒ Estimate the (latent) topological structure from observations
  - ⇒ Ex. identify high-order interactions in brain networks

Machine Learning with Topologies

Core objective: learn representations from topological structures and signals

- Leverage structure invariants
- Learn expressive mapping from data

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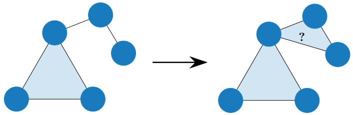
- Closure prediction
  - ⇒ Predict missing simplexes from the structure
  - ⇒ E.g., predict the creation of social groups



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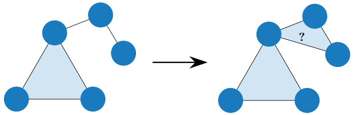
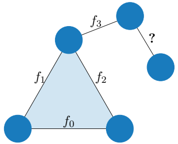


- Simplex classification
  - ⇒ Estimate the signal label on a simplex
  - ⇒ Ex. classify traffic flows



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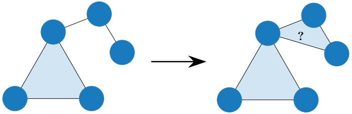

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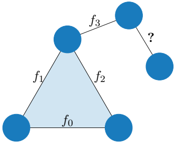
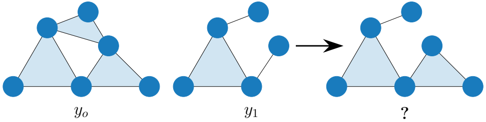


- Simplicial complex classification
  - ⇒ Predict the label of a simplicial complex
  - ⇒ Ex. classify functionality of a molecule



Topological Structures

Graphs are Basic Topological Structures

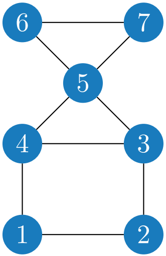
- Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
  - $\Rightarrow$ A set $\mathcal{V}$ of vertices or nodes $\mathcal{V} = \{1, 2, 3, 4, 5, 6, 7\}$ in the figure
  - $\Rightarrow$ Connected by a set $\mathcal{E} = \{(i, j)\}$ of edges or links
- Representations
  - $\Rightarrow$ adjacency matrix $\mathbf{A}$

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

  - $\Rightarrow$ incidence matrix $\mathbf{B_1}$

$$\mathbf{B_1} = \left. \overbrace{\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}}^{\text{edges}} \right\} \text{nodes}$$

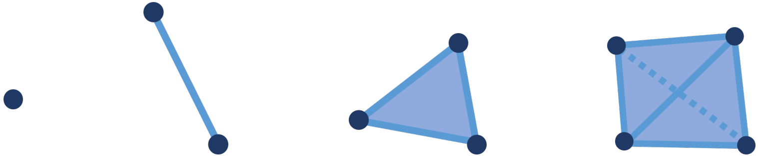
  - $\Rightarrow$ Graph Laplacian: $\mathbf{L} = \mathbf{B}_1 \mathbf{B}_1^\top$



Simplex

Definition. Given a finite set of vertices $\mathcal{V}$, a k-simplex $\mathcal{S}^k$ is an unordered subset containing $k+1$ elements $\{v_0, \ldots, v_{K+1}\}$.

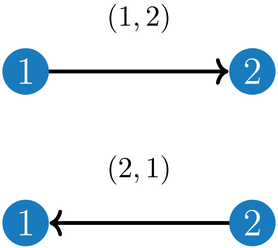
- Embedded in the Euclidean space + relationship to express closeness



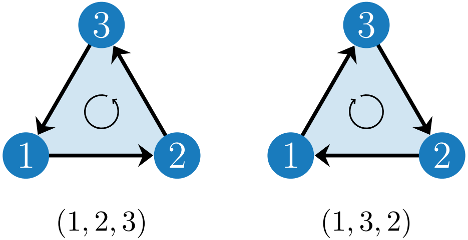
0-simplex 1-simplex 2-simplex 3-simplex

Orientations

- An oriented simplicial complex:
  - $\Rightarrow$ has a predetermined order of vertices
  - $\Rightarrow$ visualized as a walk on the simplex in this order



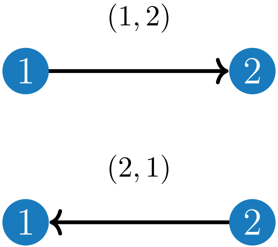
Orientations of the edge $\{1, 2\}$: $(1, 2)$ vs $(2, 1)$



Orientaions of the triangle $\{1, 2, 3\}$: $(1, 2, 3)$ vs $(1, 3, 2)$

- Ordered simplices are represented using (), and unoriented sets using {}

Orientations



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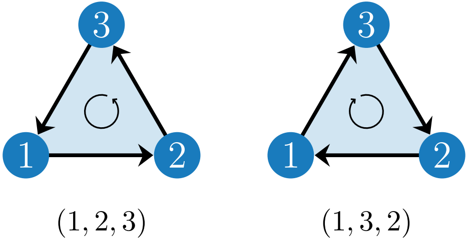

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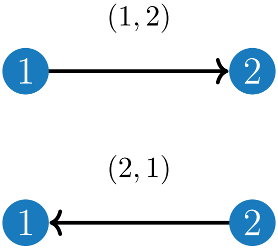

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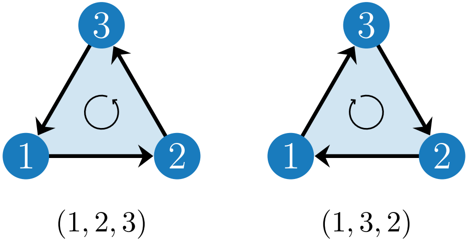
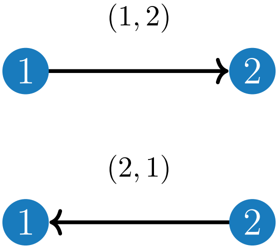


Orientaions of the triangle $\{1, 2, 3\}$: $(1, 2, 3)$ vs $(1, 3, 2)$

- With a certain starting node, each k-simplex ($k > 0$), has two orientations.

Orientations



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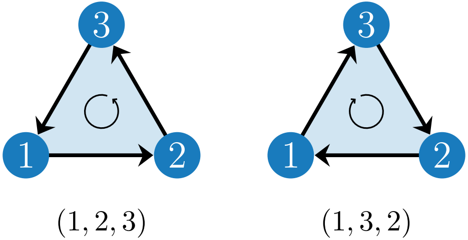

Orientaions of the triangle $\{1, 2, 3\}$: $(1, 2, 3)$ vs $(1, 3, 2)$

- With a certain starting node, each k-simplex ($k > 0$), has two orientations.
- We can then represent each of these two classes:

$\Rightarrow$

$$123 := \{(1,2,3), (2,3,1), (3,1,2)\} \quad \leftarrow \text{Even permutations}$$
$$132 := \{(1,3,2), (2,1,3), (3,2,1)\} \quad \leftarrow \text{Odd permutations}$$

Simplicial Complex

Definition. A simplicial complex of order K, $\mathcal{X}^K$ is a finite collection of all $k-$simplices $\mathcal{S}^k$ for $k = 0, 1, \ldots, K$ under the inclusion property

$$\text{for any } \mathcal{S}^k \in \mathcal{X}^K, \quad \text{all subsets } \mathcal{S}^{k-1} \subset \mathcal{S}^k \text{ belong to the simplicial complex } \mathcal{S}^{k-1} \in \mathcal{X}^K$$

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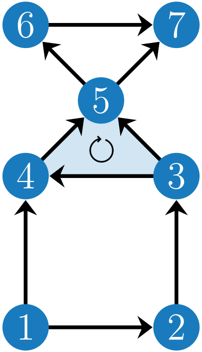
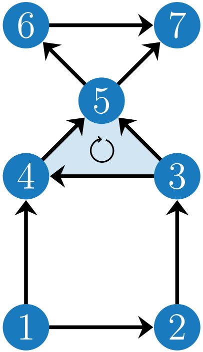
- Example: Simplicial complex (SC) $\mathcal{X}^2$ of order $K = 2$
  - $\Rightarrow$ 0-simplices $\mathcal{S}^0$: 7 nodes $\{1, \ldots, 7\}$
  - $\Rightarrow$ 1-simplices $\mathcal{S}^1$: 9 edges $\{(1,2), (2,3), \ldots.(6,7)\}$
  - $\Rightarrow$ 2-simplices $\mathcal{S}^2$: 1 filled triangle $\{(3,4,5)\}$
- Inclusion property:
  - $\Rightarrow$ Edge $(1,2)$ is part of SC only if both nodes $\{1\}$, $\{2\}$ belong to it
  - $\Rightarrow$ Filled triangle $\{(3,4,5)\}$ is part of SC only if all edges $(3,4), (4,5), (3,5)$ belong to it



Orientation in Simplicial Complexes

- We can orient a simplicial complex easily:
  - $\Rightarrow$ by defining a global order of nodes in the simplicial complex
  - $\Rightarrow$ this global order can be used to define the orientation of any simplex
- Convention for edges:
  - $\Rightarrow$ Oriented from source to sink.
  - $\Rightarrow$ Source: node with lower label
  - $\Rightarrow$ Sink: node with higher label
- Convention for triangles:
  - $\Rightarrow$ Similarly, oriented following the increasing order of node labels.



Algebraic Representation

Incidence matrices $\mathbf{B}_k$ establish incident relationships between $k-$simplices and $k+1-$simplices

- Examples of vertical relationships (figure)
  - $\Rightarrow$ $\mathbf{B}_1$: node-to-edge incidence matrix
  - $\Rightarrow$ $\mathbf{B}_2$: edge-to-triangle incidence matrix
  - $\Rightarrow$ $\mathbf{B}_k \mathbf{B}_{k+1} = \mathbf{0}$

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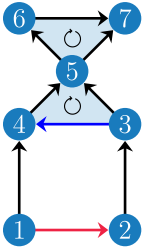


- Orientation of the triangle and edge coincide $\Rightarrow +1$
  - $\Rightarrow$ Otherwise, -1

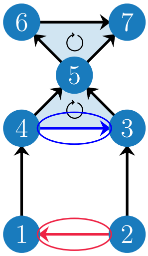
Orientations in Incidence Matrices

- What happens when we change the orienation of two edges?

Orientations in Incidence Matrices

- What happens when we change the orienation of two edges?

$$\mathbf{B}_1 = \left.\overbrace{\begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}}^{\text{edges}}\right\}\text{nodes} \quad \mathbf{B}_2 = \left.\overbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -1 & 0 \\ 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}}^{\text{triangles}}\right\}\text{edges}$$



- In $\mathbf{B}_1$, swap the signs in the column corresponding to the changed edges.
- In $\mathbf{B}_2$, the edge orientation opposes the orienation of the triangle!

  $\Rightarrow$ becomes -1

Algebraic Representation

- Hodge Laplacians: describe the structure of the whole $K-$simplicial complex

$$\mathbf{L}_0 = \mathbf{B}_1\mathbf{B}_1^\top \longleftarrow \text{graph Laplacian}$$

$$\mathbf{L}_1 = \mathbf{B}_1^\top\mathbf{B}_1 + \mathbf{B}_2\mathbf{B}_2^\top$$

$$\mathbf{L}_k = \mathbf{B}_k^\top\mathbf{B}_k + \mathbf{B}_{k+1}\mathbf{B}_{k+1}^\top \quad k = 2, \ldots, K-1$$

$$\mathbf{L}_K = \mathbf{B}_K^\top\mathbf{B}_K$$

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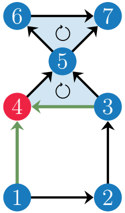
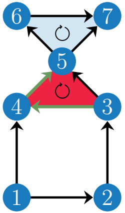
\end{aligned}
$$

- Intermediate Laplacians encode both lower and upper neighborhood relations
  - $\Rightarrow$ Lower Laplacian: $\mathbf{L}_{k,\ell} = \mathbf{B}_k^\top\mathbf{B}_k$ for lower adjacencies: Edges (3, 4) and (1, 4) are lower adjacent
  - $\Rightarrow$ Up Laplacian: $\mathbf{L}_{k,u} = \mathbf{B}_{k+1}\mathbf{B}_{k+1}^\top$ for upper adjacencies: Edge (3, 4) and (4, 5) are upper adjacent





Cells

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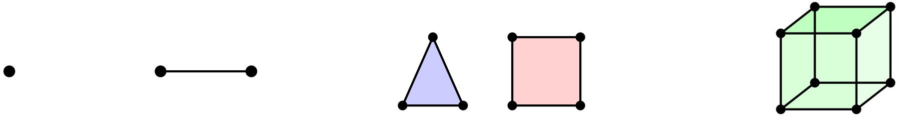
- Unlike simplices, a k-cell is not uniquely defined by a set of vertices, but by its boundary (i.e. how its $(k-1)$- cells are arranged).



0-cell (nodes)

1-cell (edges: structured collection of nodes)

2-cells (polygons: structured collections of edges)

3-cells (volumes: structured collections of polygons)

Cell Complex

Definition. A cell complex of order K, $\mathcal{Y}^K$ is a finite collection of all k-cells $\mathcal{C}^k$ for $k = 0, 1, \ldots, K$ together with boundary relations between cells of consecutive dimensions.

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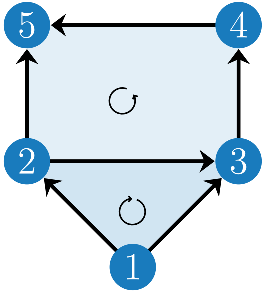
- Example: Cell complex $\mathcal{Y}^2$ of order $K = 2$
  - $\Rightarrow$ 0-cells $\mathcal{C}^0$: 5 nodes $\{1, \dots, 5\}$
  - $\Rightarrow$ 1-cells $\mathcal{C}^1$: 6 edges $\{(1,2), (2,3), \dots .(4,5)\}$
  - $\Rightarrow$ 2-cells $\mathcal{C}^2$: 2 polygons: 1 filled triangle $\{(1,2,3)\}$ and 1 filled square {(2,3,4,5)}.
- No requirement to satisfy the inclusion property!
  - $\Rightarrow$ A 2-cell $(2, 3, 4, 5)$ can exist without triangles $(2, 3, 4)$, $(2, 3, 5)$ etc.
  - $\Rightarrow$ Every cell must still have its boundaries: a polygon cannot exist without its edges



Algebraic Representation

Incidence matrices $\mathbf{B}_k$ establish incident relationships between k-cells and $k+1$-cells.

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- Examples of vertical relationships (figure)
  - ⇒ $\mathbf{B}_1$: node-to-edge incidence matrix
  - ⇒ $\mathbf{B}_2$: edge-to-polygon incidence matrix
  - ⇒ $\mathbf{B}_k\mathbf{B}_{k+1} = \mathbf{0}$: ensures cells have all their boundaries

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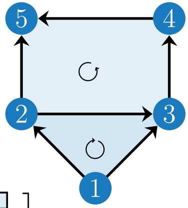

$$\mathrm{B}_1 = \begin{array}{c|cccccc} & (1,2) & (1,3) & (2,3) & (3,4) & (4,5) & (2,5) \\ \hline 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & -1 & 0 & 0 & -1 \\ 3 & 0 & 1 & 1 & -1 & 0 & 0 \\ 4 & 0 & 0 & 0 & 1 & -1 & 0 \\ 5 & 0 & 0 & 0 & 0 & 1 & 1 \end{array}$$

$$\mathrm{B}_2 = \begin{array}{c|cc} & \triangle & \square \\ \hline (1,2) & 1 & 0 \\ (1,3) & -1 & 0 \\ (2,3) & 1 & 1 \\ (3,4) & 0 & 1 \\ (4,5) & 0 & 1 \\ (2,5) & 0 & -1 \end{array}$$

Cell Complex

Cell complex provides a more efficient way to capture higher-order relations by removing the restriction to inclusion property

$$\mathbf{L}_0 = \mathbf{B}_1\mathbf{B}_1^\top$$
$$\mathbf{L}_1 = \mathbf{B}_1^\top\mathbf{B}_1 + \mathbf{B}_2\mathbf{B}_2^\top$$
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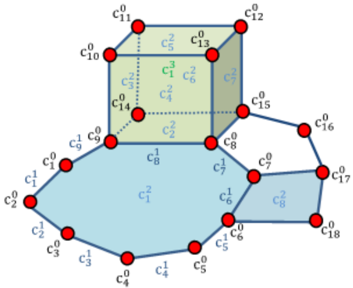

- Ex. Matrix $\mathbf{B}_2$ contains now edge-to-polygon incidences

Hypergraphs

Hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ consists of vertices $\mathcal{V}$ and hyperedges $\mathcal{E}$ (subsets of $\mathcal{V}$).

- Advantages
  - $\Rightarrow$ General structure: hyperedges can connect any number of vertices
  - $\Rightarrow$ Flexibility: high degrees-of-freedom to model structure
  - $\Rightarrow$ Special case: reduces to a graph if all hyper-edges connect exactly two vertices.
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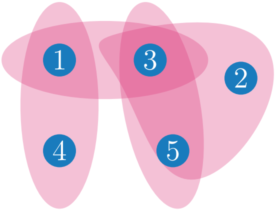
  - ⇒ Useful when only relationships are needed

- Limitations
  - ⇒ High computational complexity
  - ⇒ Not a unique representation
  - ⇒ Lacks an algebraic representation
  - ⇒ Cannot capture hierarchical information in data



- Here we will use complexes, predominantly simplicial complexes of order $K = 2$

Topological Signals

Simplicial Signals

Definition. A simplicial signal is a mapping from the simplex set the real domain $f : \mathcal{S}^k \to \mathbb{R}$.

- Simplicial signals $\Rightarrow$ Associate a value to each simplex

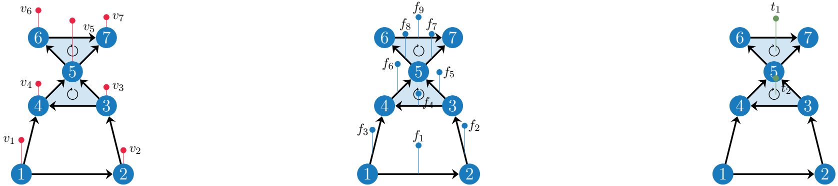
  $\Rightarrow$ collect in vector $\mathbf{x}^k = \left[x_1^k, \ldots, x_{N_k}^k\right]^\top \in \mathbb{R}^{N_k}$

- Examples:

  $\Rightarrow$ Node signals: $\mathbf{x}^0 := \mathbf{v} \in \mathbb{R}^{N_0}$ measurements of temperature, water pressure

  $\Rightarrow$ Edge signals: $\mathbf{x}^1 := \mathbf{f} \in \mathbb{R}^{N_1}$ Measurement of water flow, traffic flow

  $\Rightarrow$ Triangle signals: $\mathbf{x}^2 := \mathbf{t} \in \mathbb{R}^{N_2}$ EEG measurements of three voxel interaction



Node signals   Edge signals   Triangle signals

Signal Mappings

- Incidence Matrices $\mathbf{B}_k$
  - $\Rightarrow$ established relationships among simplices on the k^{th} and $k+1^{th}$ level.
  - $\Rightarrow$ can also be seen as linear operators that map signals on the k^{th} level to the $k+1^{th}$ level.

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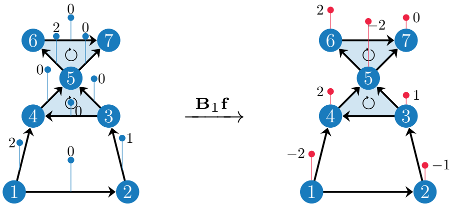
- $\mathbf{B}_1$: Edge to node signal mappings
  - $\Rightarrow$ Let us start with an edge signal, $\mathbf{f}$.
  - $\Rightarrow$ $\mathbf{B}_1\mathbf{f} = \tilde{\mathbf{v}}$, where $\tilde{\mathbf{v}}$ is now a signal on the nodes!
- Example:



Edge signal, $\mathbf{f}$ Node signal, $\tilde{\mathbf{v}}$

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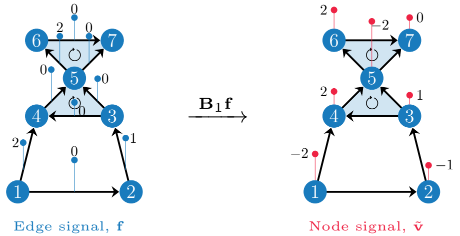
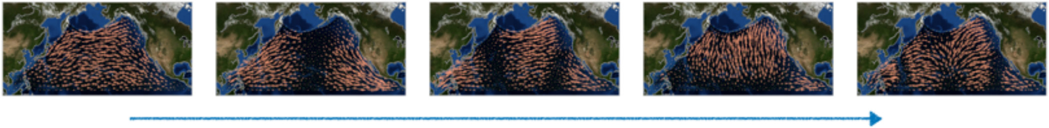


- $\tilde{\mathbf{v}}$ has a special meaning:
  - $\Rightarrow$ Every entry tells the net in-flow or out-flow on that node.
  - $\Rightarrow$ $\tilde{\mathbf{v}}$ is the divergence of $\mathbf{f}$.
  - $\Rightarrow$ $\mathbf{B}_1$: divergence operator

Signal Mappings

- The energy of the vector $||\mathbf{B}_1\mathbf{f}||_2^2$ gives the total divergence.
  - $\Rightarrow$ Larger the value of $||\mathbf{B}_1\mathbf{f}||_2^2$, larger the divergence of the edge signals.



Edge signals with increasing divergence, $||\mathbf{B}_1\mathbf{f}||_2^2$

Signal Mappings

- $\mathbf{B}_1^\top$: Node to edge signal mappings
  - $\Rightarrow$ Let us start with an node signal, $\mathbf{v}$.
  - $\Rightarrow$ $\mathbf{B}_1^\top \mathbf{v} = \tilde{\mathbf{f}}$, where $\tilde{\mathbf{f}}$ is now a signal on the edges!

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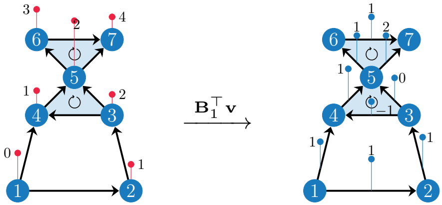

Node signal, $\mathbf{v}$ | Edge signal, $\tilde{\mathbf{f}}$

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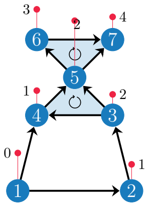
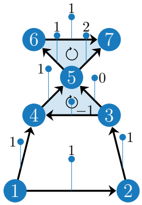


Node signal, $\mathbf{v}$

$\xrightarrow{\mathbf{B}_1^\top \mathbf{v}}$

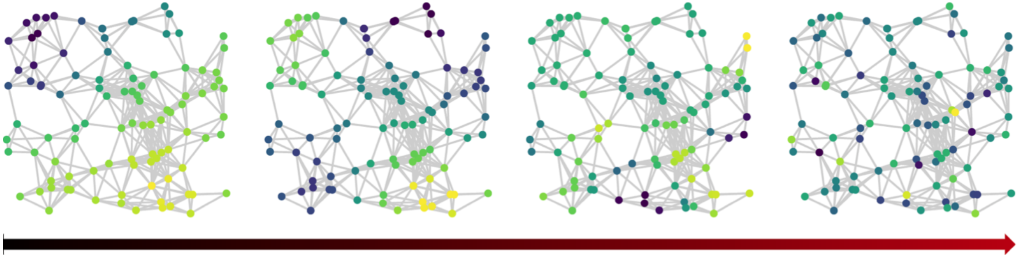


Edge signal, $\tilde{\mathbf{f}}$

- $\tilde{\mathbf{f}}$ has a special meaning:
  - $\Rightarrow$ Every entry of $\tilde{\mathbf{f}}$ tells us the pair-wise node signal differences belonging to that edge.
  - $\Rightarrow$ $\tilde{\mathbf{f}}$ is the gradient signal of $\mathbf{v}$.
  - $\Rightarrow$ $\mathbf{B}_1^\top$: gradient operator.

Signal Mappings

- The energy of the vector $||\mathbf{B}_1^\top \mathbf{v}||_2^2$ gives the total pair-wise differences (graph signal smoothness!)
  - $\Rightarrow$ Larger the value of $||\mathbf{B}_1^\top \mathbf{v}||_2^2$, larger the variability of the node signals across edges.



Node signals with increasing variation across edges, $||\mathbf{B}_1^\top \mathbf{v}||_2^2$

fig. credit Hansen et al.,

Signal Mappings

- $\mathbf{B}_2$: Triangle to edge signal mappings
  - $\Rightarrow$ Let us start with a triangle signal, $\mathbf{t}$.
  - $\Rightarrow$ $\mathbf{B}_2\mathbf{t} = \tilde{\mathbf{f}}$, where $\tilde{\mathbf{f}}$ is a signal on the edges.

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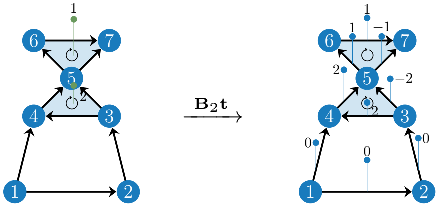

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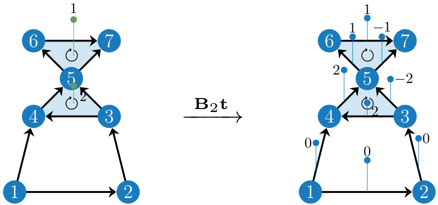

Triangle signal, $\mathbf{t}$

Edge signal, $\tilde{\mathbf{f}}$

- $\tilde{\mathbf{f}}$ has a special meaning:
  - $\Rightarrow$ local flow within a triangle
  - $\Rightarrow$ $\tilde{\mathbf{f}}$ has zero net flow at each node. Thus $\tilde{\mathbf{f}}$ curls around a triangle.
  - $\Rightarrow$ $\tilde{\mathbf{f}}$ is called a curl flow,
  - $\Rightarrow$ $\mathbf{B}_2$ is called the curl adjoint.

Signal Mappings

- $\mathbf{B}_2^\top$: Edge to triangle signal mappings
  - $\Rightarrow$ Start with an edge signal, $\mathbf{f}$.
  - $\Rightarrow$ $\mathbf{B}_2^\top \mathbf{f} = \tilde{\mathbf{t}}$, where $\tilde{\mathbf{t}}$ is a signal on the triangles.

Signal Mappings

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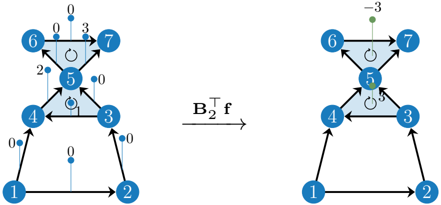

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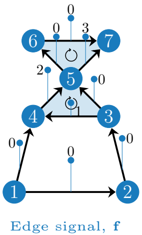
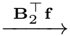
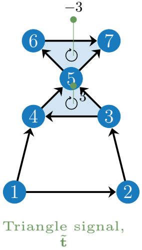






- $\tilde{\mathbf{t}}$ has a special meaning:

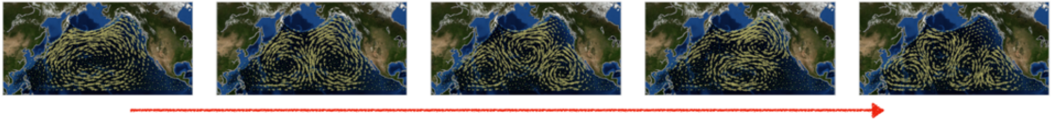
  $\Rightarrow$ Every entry of $\tilde{\mathbf{t}}$: signed sum of the edge signals of that triangle.

  $\Rightarrow$ $\tilde{\mathbf{t}}$ represents the extent of rotation of the edge signals around a triangle, i.e., the curl.

  $\Rightarrow$ $\mathbf{B}_2^\top$ is thus called the curl operator.

Signal Mappings

- The energy of the vector $||\mathbf{B}_2^\top \mathbf{f}||_2^2$ gives the total curl of the edge signal.
  $\Rightarrow$ Larger the value of $||\mathbf{B}_2^\top \mathbf{f}||_2^2$, larger the rotation of the edge signals.



Edge signals with increasing rotation (curl), $||\mathbf{B}_2^\top \mathbf{f}||_2^2$

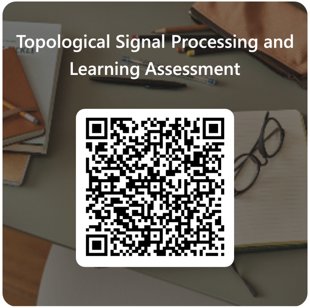
Learning Activity: Divergence vs Curl in Edge Flows

A simplicial complex models a flow network. Node signals represent pressure, edge signals represent flow, and triangle signals represent local circulation. You observe an edge flow signal $\mathbf{f} \in \mathbb{R}^{N_1}$ and want to quantify whether it contains mainly sources/sinks or circulation.

- Q4.1: Which expression correctly measures the total divergence and the total curl of an edge flow signal $\mathbf{f}$, using incidence matrices $\mathbf{B}_1$ and $\mathbf{B}_2$?

- Instructions
  - ⇒ Scan the QR code to open the activity
  - ⇒ Work individually first
  - ⇒ Then discuss your answer in pairs or groups of 3
  - ⇒ Submit your final answer in Microsoft Forms

- Logistics
  - ⇒ Time: 10 min
  - ⇒ Goal: Identify the correct operators and explain their physical meaning
  - ⇒ Product: One submitted response per student + justification to your peers



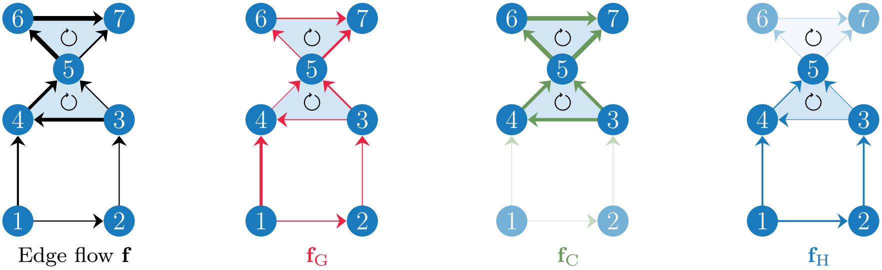
https://forms.office.com/

Hodge Decomposition

$$\mathbb{R}^{N_1} \equiv \underbrace{\mathrm{im}\left(\mathbf{B}_1^\top\right)}_{\text{Gradient}} \oplus \underbrace{\ker\left(\mathbf{L}_1\right)}_{\text{Harmonic}} \oplus \underbrace{\mathrm{im}\left(\mathbf{B}_2\right)}_{\text{curl}}$$

- An edge flow signal can be decomposed as

$$\mathbf{f} = \mathbf{f}_\mathrm{G} + \mathbf{f}_\mathrm{C} + \mathbf{f}_\mathrm{H}$$



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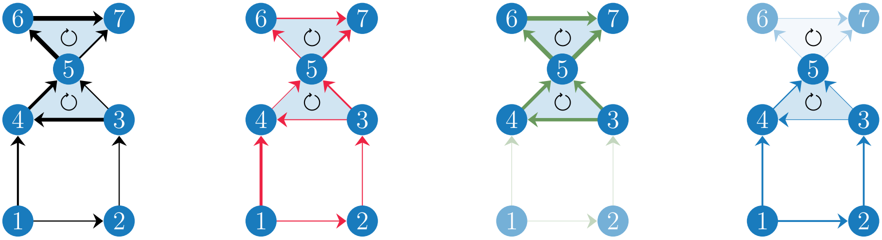

Edge flow $\mathbf{f}$ $\qquad$ $\mathbf{f}_{\mathrm{G}}$ $\qquad$ $\mathbf{f}_{\mathrm{C}}$ $\qquad$ $\mathbf{f}_{\mathrm{H}}$

$\Rightarrow$ Gradient flow $\mathbf{f}_{\mathrm{G}}$: $= \mathbf{B}_1^\top \mathbf{v}$ difference of node signals, zero flow-sum along triangles

Hodge Decomposition

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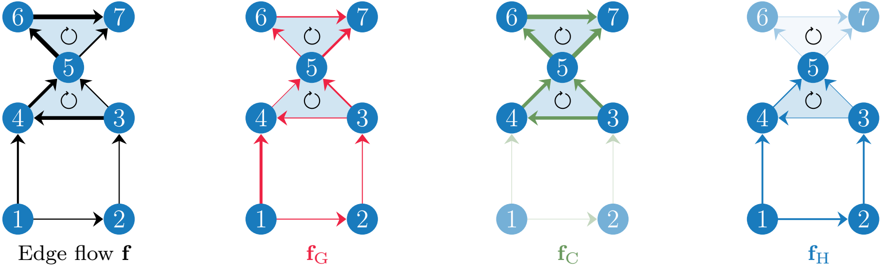


$\Rightarrow$ Gradient flow $\mathbf{f}_{\mathrm{G}}$: $= \mathbf{B}_1^\top \mathbf{v}$ difference of node signals, zero flow-sum along triangles

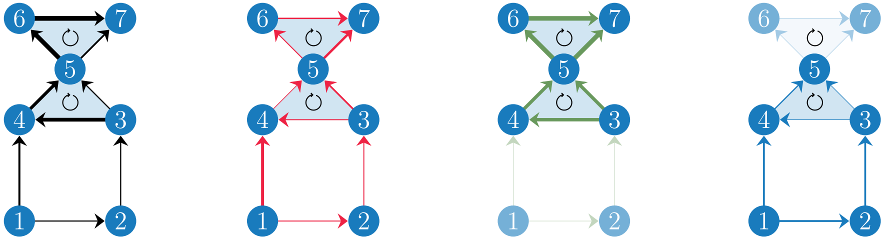
$\Rightarrow$ Curl flow $\mathbf{f}_{\mathrm{C}}$: $= \mathbf{B}_2 \mathbf{t}$ local flows circulating around triangles, in-flow on each node equals out-flow

Hodge Decomposition

$$\mathbb{R}^{N_1} \equiv \underbrace{\operatorname{im}\left(\mathbf{B}_1^\top\right)}_{\text{Gradient}} \oplus \underbrace{\ker\left(\mathbf{L}_1\right)}_{\text{Harmonic}} \oplus \underbrace{\operatorname{im}\left(\mathbf{B}_2\right)}_{\text{curl}}$$

- An edge flow signal can be decomposed as

$$\mathbf{f} = \mathbf{f}_\mathrm{G} + \mathbf{f}_\mathrm{C} + \mathbf{f}_\mathrm{H}$$



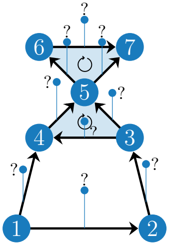
$\Rightarrow$ Gradient flow $\mathbf{f}_\mathrm{G}$: $= \mathbf{B}_1^\top \mathbf{v}$ difference of node signals, zero flow-sum along triangles

$\Rightarrow$ Curl flow $\mathbf{f}_\mathrm{C}$: $= \mathbf{B}_2 \mathbf{t}$ local flows circulating around triangles, in-flow on each node equals out-flow

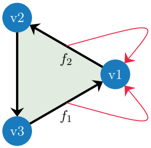
$\Rightarrow$ Harmonic flow $\mathbf{f}_\mathrm{H}$: global circulating flow

Topological Signal Regularizers

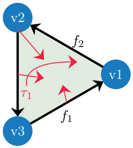
$$\underset{\hat{\mathbf{f}}\in\mathbb{R}^{N_1}}{\operatorname{argmin}}\ \|\mathbf{P}(\hat{\mathbf{f}}-\mathbf{y})\|_2^2+\alpha\|\mathbf{B}_1\hat{\mathbf{f}}\|_q^q+\beta\|\mathbf{B}_2^\top\hat{\mathbf{f}}\|_p^p$$



- Optimization problem
  - ⇒ $\mathbf{y}$ is the measurements with noise and missing values
  - ⇒ $\mathbf{P}$ indicates the position of the missing values
  - ⇒ $\alpha\|\mathbf{B}_1\hat{\mathbf{f}}\|_q^q$ reduces the energy of the divergence
  - ⇒ $\beta\|\mathbf{B}_2^\top\hat{\mathbf{f}}\|_p^p$ reduces the energy of the curl



$\mathbf{B}_1\mathbf{f}$



$\mathbf{B}_2^\top\mathbf{f}$

Dirac Operator

- To process topological signals across different dimensions jointly, we define the joint signal vector as:

$$\mathbf{x} = [\mathbf{v}^\top, \mathbf{f}^\top, \mathbf{t}^\top]^\top$$

$\Rightarrow$ $\mathbf{v}$, $\mathbf{f}$, and $\mathbf{t}$ represent the node, edge, and triangle signals, respectively.

- The Dirac operator $\mathbf{D}$ is defined as the sum of the lower and upper Dirac operators ($\mathbf{D} = \mathbf{D}_l + \mathbf{D}_u$):

$$\mathbf{D} = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix}$$

- where the lower ($\mathbf{D}_l$) and upper ($\mathbf{D}_u$) Dirac operators are explicitly defined as:

$$\mathbf{D}_l = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{D}_u = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix}$$

- The Dirac operator inherently couples adjacent dimensions which process all topological signals.

Joint Application of the Dirac Operator

- Applying the Dirac operator to the joint signal $\mathbf{x}$ simultaneously processes all topological orders:

$$\mathbf{D}\mathbf{x} = \begin{bmatrix} \mathbf{B}_1\mathbf{f} \\ \mathbf{B}_1^\top\mathbf{v} + \mathbf{B}_2\mathbf{t} \\ \mathbf{B}_2^\top\mathbf{f} \end{bmatrix}$$

- This operation jointly captures three distinct topological interactions:
  - ⇒ The divergence of edge flows ($\mathbf{B}_1\mathbf{f}$).
  - ⇒ The sum of node gradients and triangle curls ($\mathbf{B}_1^\top\mathbf{v} + \mathbf{B}_2\mathbf{t}$).
  - ⇒ The curl of edge flows ($\mathbf{B}_2^\top\mathbf{f}$).
- Furthermore, applying the Dirac operator twice yields the block-diagonal Hodge Laplacian matrix $\boldsymbol{\mathcal{L}}$:

$$\mathbf{D}^2 = \boldsymbol{\mathcal{L}} = \begin{bmatrix} \mathbf{L}_0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{L}_2 \end{bmatrix}$$

Joint Signal Reconstruction: Dirac Regularization

- We aim to reconstruct the complete joint signal $\hat{\mathbf{x}}$ from corrupted or partial joint measurements $\mathbf{y}$.
- Instead of regularizing nodes, edges, and triangles separately, we process them jointly.
- The regularized optimization problem with a Dirac ℓ_2-norm is formulated as:

$$\underset{\hat{\mathbf{x}}}{\operatorname{argmin}} \|\mathbf{P}(\hat{\mathbf{x}} - \mathbf{y})\|_2^2 + \gamma \|\mathbf{D}\hat{\mathbf{x}}\|_2^2$$

- This formulation elegantly enforces global smoothness across all signal dimensions since:

$$\|\mathbf{D}\hat{\mathbf{x}}\|_2^2 = \hat{\mathbf{x}}^\top \mathbf{D}^2 \hat{\mathbf{x}} = \hat{\mathbf{x}}^\top \boldsymbol{\mathcal{L}} \hat{\mathbf{x}}$$

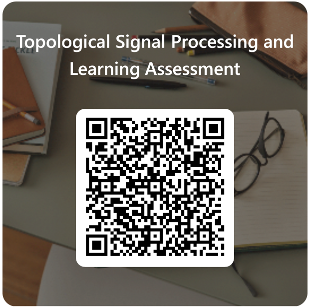
Learning Activity: Dirac Operator as Joint Coupling Mechanism

To jointly process all topological signals, we use the Dirac operator, which couples only adjacent dimensions: nodes interact with edges, and edges interact with triangles.

- Q4.2: When applying the Dirac operator to the joint signal, the output consists of three blocks: a node-level output, an edge-level output, and a triangle-level output. Which option correctly describes these three outputs?

- Instructions
  - ⇒ Scan the QR code to open the activity
  - ⇒ Work individually first
  - ⇒ Then discuss your answer in pairs or groups of 3
  - ⇒ Submit your final answer in Microsoft Forms

- Logistics
  - ⇒ Time: 10 min
  - ⇒ Goal: Understand how the Dirac operator couples node, edge, and triangle signals
  - ⇒ Product: One submitted response per student + justification to your peers



https://forms.office.com/

Further Reading

- Applications of Higher-Order Networks:

  ⇒ Owen, L.L.W., Chang, T.H. & Manning, J.R. High-level cognition during story listening is reflected in high-order dynamic correlations in neural activity patterns. Nat Commun 12, 5728 (2021). https://doi.org/10.1038/s41467-021-25876-x

  ⇒ Burgio, G., St-Onge, G. & Hébert-Dufresne, L. Characteristic scales and adaptation in higher-order contagions. Nat Commun 16, 4589 (2025). https://doi.org/10.1038/s41467-025-59777-0

  ⇒ Murgas, K.A., Saucan, E. & Sandhu, R. Hypergraph geometry reflects higher-order dynamics in protein interaction networks. Sci Rep 12, 20879 (2022). https://doi.org/10.1038/s41598-022-24584-w

  ⇒ Hansen et al., Power flow balancing with decentralized graph neural networks, IEEE Trans. on Power Systems, 2022

- Foundamentals:

  ⇒ Bick, C., Gross, E., Harrington, H. A., and Schaub, M. T. (2023). What are higher-order networks?. SIAM review, 65(3), 686-731.

  ⇒ Hoppe, Josef, Vincent P. Grande, and Michael T. Schaub. "Don't be Afraid of Cell Complexes! An Introduction from an Applied Perspective." arXiv preprint arXiv:2506.09726 (2025).

  ⇒ Papillon, et al., Beyond Euclid: An illustrated guide to modern machine learning with geometric, topological, and algebraic structures. Machine Learning: Science and Technology, 6(3), 031002.

  ⇒ Liu, C., Leus, G., and Isufi, E. (2023). Unrolling of simplicial elasticnet for edge flow signal reconstruction. IEEE Open Journal of Signal Processing, 5, 186-194.